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Amphiphiles in aqueous solution: well beyond a soap bubble

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Owing to their ''dual'' affinity, amphiphiles self-assemble in water to form different kinds of nanoscale multimolecular assemblies ranging from simple micelles and vesicles to highly organized fibers, helices and tubes. In this *tutorial review* the aggregates formed in water by head/tail amphiphiles are revisited and discussed from the point of view of supramolecular chemistry with a focus on their structure and recognition abilities. Their applications in materials chemistry, as soft templates for inorganic nanostructures, as well as in biological and medicinal chemistry are also illustrated. Special attention is paid to highlight intriguing aspects, for example the control of morphology and chirality, their modulation by experimental parameters and chiral symmetry breaking.

Key learning points

(1) Organized supramolecular assemblies of amphiphiles: control of their morphology and chirality.
(2) Molecular and chiral recognition in aggregates of amphiphiles (biomembrane models).
(3) Amphiphile aggregates as templates for inorganic structures.
(4) Biological/medicinal applications of amphiphiles.

1. Introduction

The word amphiphile comes from the Greek roots: *αμφις, amphis*, which means ''both'', ''double'' and *φιλία, philia*, which expresses affinity. It is the term used to indicate compounds exhibiting a dual affinity, *i.e.* possessing both hydrophilic (water-loving) and lipophilic (fat-loving) properties.

A typical amphiphilic molecule consists of a polar hydrophilic group, usually called the *head*, joined to a nonpolar hydrophobic moiety called the *tail* (Fig. 1). However, the number of the polar head(s) and of the hydrophobic tail(s) and the kind of connection among them can vary, which leads to different classes of amphiphiles (see below). Typically, the head consists of a charged or uncharged polar group, like phosphate, sulfate, ammonium, amino acid, peptide or sugar. The hydrophobic tail usually consists of a hydrocarbon chain, which may be saturated or unsaturated, linear or branched.

Common amphiphiles are surfactants, such as detergents, soaps and fatty acids, or diglycerides, such as phospholipids, the latter being the main components of biomembranes. Surfactants (short for *surface active agents*) are amphiphilic molecules that exhibit interfacial activity usually consisting in their ability to lower the surface tension of a liquid or the interfacial tension between two liquids of different polarity due to their property of accumulating at the interface. Due to their characteristics, they are widely used as detergents, wetting agents, dispersants, emulsifiers, foaming agents, *etc.* and are often labeled according to their use. It is worth remarking that not all amphiphiles are surface active molecules but more importantly, they create surfaces.[1] In fact, only the amphiphiles with more or less equilibrated hydrophilic and lipophilic tendencies are likely to migrate to the surface or interface.

Amphiphiles are usually classified according to different criteria, including the charge of the head group upon its dissociation in water (often used for surfactants), the number and kind of connection of polar head(s)/hydrophobic tail(s) (Fig. 1) and the chemical nature of the polar head. Regarding the head group charge, surfactants are classified in ionic (cationic or anionic), non-ionic, zwitterionic and amphoteric surfactants (whose charge changes depending on pH). Based on the number and kind of connection of polar head(s)/hydrophobic tail(s) amphiphiles are classified as:

– conventional single head/single tail amphiphiles;

– bolaamphiphiles, which contain two hydrophilic heads connected by a hydrophobic skeleton (*e.g.* one or two alkyl

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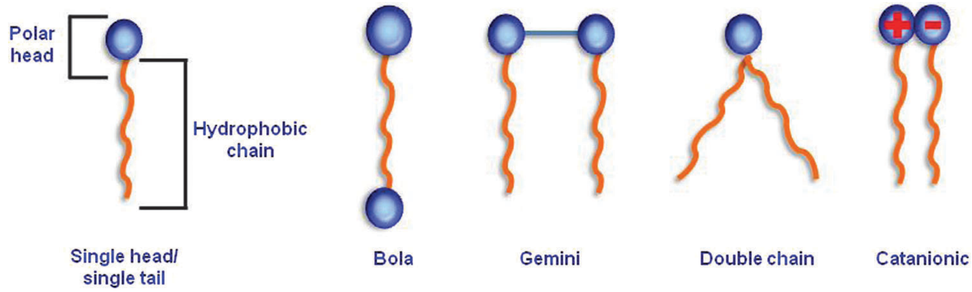

Fig. 1 Amphiphile classification.

chains, a steroid, or a porphyrin). They owe their name to the "bolas" the weapons used by the South American gauchos to hunt;

– gemini or "dimeric" surfactants made up of two hydrocarbon tails and two ionic groups linked by a spacer. They feature unique properties among surfactants such as an extremely stronger surface activity and lower *critical micelle concentration* (*cmc*, *i.e.* minimum concentration at which aggregation occurs) than conventional amphiphiles;

– double and triple chain amphiphiles;

A special kind are catanionic amphiphiles, which result from the 1 : 1 mixing of oppositely charged surfactants acting as counterions to each other.

Beyond the more conventional head/tail amphiphiles, other species often possess amphiphilic behaviour, such as peptides and proteins, amphiphilic polymers (*e.g.* block copolymers), "facial amphiphiles", in which the hydrophilic and hydrophobic moieties are located on two opposite faces of the molecule, and amphiphilic dyes.

As a consequence of their double affinity, amphiphilic molecules self-organize in water to form nanoscale polymolecular assemblies in which the hydrophobic moieties are buried into the aggregate, shielded from the contact with water, while the hydrophilic groups remain exposed at the surface ensuring solubility.

As proposed by Tanford,[2] two opposing forces are responsible for the formation of such assemblies. The *hydrophobic effect* drives the segregation of the hydrocarbon chains in water, thus providing the impetus for self-organization. On the other hand,

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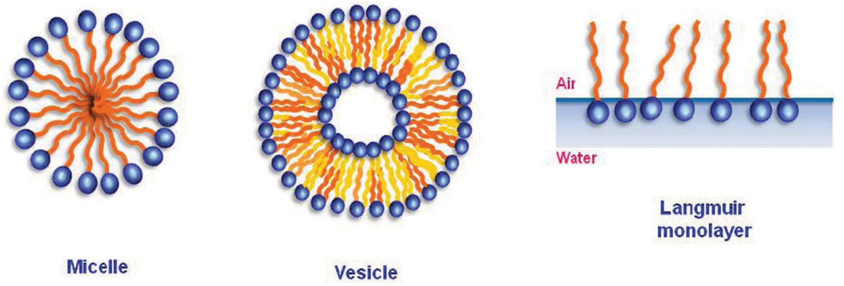


Fig. 2 Amphiphile assemblies.

repulsive forces (electrostatic and/or hydratation) and steric effects between the polar heads prevent the formation of large three-dimensional crystals and hence phase separation.[3]

The self-aggregation of amphiphilic molecules has long been known to yield a rich variety of assemblies, depending on the molecular structure of the amphiphile (interplay between the hydrophilic–hydrophobic balance and geometric packing constraints) and on experimental conditions, such as concentration, temperature, pH and ionic strength.[4] Among the most common self-assemblies, of colloid dimensions, formed in water by amphiphilic molecules are micelles and vesicles (Fig. 2), also referred to as association colloids.

Micelles are small, ill-defined assemblies of 50–100 unimers that feature an external polar surface made of the hydrophilic head groups and a more or less disordered hydrophobic core formed by the amphiphile tails, usually regarded as a small volume of liquid hydrocarbon. They are thermodynamic equilibrium systems with a loose dynamic structure in which the unimers exchange rapidly with the bulk solution. Typically, micelles are formed spontaneously and reversibly by single head/single tail surfactants (with the head substantially bigger that the tail) above *cmc*, and at temperatures above the Krafft temperature (*i.e.* the minimum temperature at which aggregation occurs). Most commonly, micelles have spherical shapes, but depending on surfactant geometry, concentration and other experimental variables, they can be ellipsoids or long cylinders (rods).[4]

Vesicles are spherical or ellipsoidal closed amphiphile bilayer structures with an internal cavity containing the aqueous solution in which they are dispersed. Typical vesicles are formed by natural phospholipids, and in such cases they are referred to as liposomes. However, vesicles are also formed by many synthetic amphiphiles and surfactants able to form bilayers (see below). With a few exceptions, vesicles are metastable in aqueous solution: energy is required to dissolve the amphiphile in water and to induce aggregation (stirring, sonication, extrusion), and eventually vesicles flocculate into hydrated multilayers. Depending on the experimental conditions of preparation, their dimensions can range from 20 nm to 10 μm and they may contain one or more concentric bilayer surfaces in an onion-like structure.

At higher amphiphile concentrations, liquid crystals (LC) can be formed. These are long-range ordered yet fluid phases in which water and amphiphile combine to form anisotropic organized structures called lyotropic mesomorphic phases (as opposed to thermotropic mesomorphs formed by heating some organic crystals). For example, as the concentration of surfactant increases, a typical phase sequence would be spherical micelles, ellipsoidal micelles, rods, hexagonal LC phase (hexagonal arrangement of long cylinders), lamellar LC phase and, eventually, reverse phases.

Among the various amphiphile assemblies, Langmuir monolayers are worthy of special mention (Fig. 2). They are monomolecular layers formed at the water–air interface by insoluble or sparingly soluble amphiphiles (both natural lipids and synthetic amphiphiles). Typically, they are obtained by spreading a dilute solution of the amphiphile in a volatile water immiscible solvent on the surface of an aqueous subphase, which fills the Langmuir trough. Molecules spread over the entire surface with the polar head oriented toward the aqueous subphase and the hydrophobic tails assuming various orientations relative to the surface plane, depending upon the available area per molecule. Two movable barriers permit compression of the monolayer, while measuring the surface pressure (π) (tension), which allows to easily control the degree of molecular packing. Thus, upon compression, the monolayer undergoes various phase transitions between gas, liquid expanded, liquid condensed and solid phases.

As stressed by F. M. Menger, it is a historical fact that colloid chemistry has been involved with self-assembly of amphiphilic molecules well before the advent of supramolecular chemistry, but even when the term supramolecular chemistry expanded to include large self-assembled polymolecular structures, like films, gels, *etc.* colloid and supramolecular chemistry still remained two widely distinct disciplines.[5] However, fortunately, in the last few decades this rather artificial distinction has begun to disappear and growing interest of supramolecular chemists in amphiphile assemblies has been emerging.

This tutorial review deals with multimolecular assemblies formed in water solution by amphiphiles reviewed and described from the point of view of supramolecular chemistry including their structure and recognition abilities. Their applications in materials science as well as in biological and medicinal chemistry are also presented. The aim is to provide insight into the fascinating field of the complex supramolecular assemblies of amphiphilic molecules, highlighting significant examples where intriguing phenomena are put in evidence (control of morphology and chirality, modulation of structure and recognition by external stimuli, chiral symmetry breaking and so on).

We constrain ourselves to canonical head(s)/tail(s) amphiphiles whose aggregation is driven by the hydrophobic effect and other structure directing interactions. Amphiphilic macromolecules such as proteins or block copolymers are beyond the scope of this review.

2. Amphiphiles in supramolecular chemistry

2.1 Organized supramolecular nanostructure: from micellar fibers to smart hydrogels

Beyond the more common micelles, vesicles and liquid crystalline mesomorphic phases, amphiphilic molecules can hierarchically self-assemble in a rich variety of more organized nanostructures such as fibers, ribbons, helices, super helices and tubes, which have attracted the interest of supramolecular chemists. This is due to the potentiality of obtaining amphiphile based self-assembled systems with designed morphologies and a less fluid and more defined structure than micellar assemblies. Thus, they can be target assemblies for the preparation of smart systems by using appropriately designed amphiphiles tailored with suitable functional groups and/or by playing with the experimental variables in order to control the aggregate morphology. The importance of morphology in relation to the function is well known in biological systems and it is an issue of growing interest in chemistry for the application in functional materials, nanochemistry and technology. In some cases, the introduction of suitable groups in the amphiphilic scaffold has allowed preparation of responsive aggregates whose morphology can be modulated by external stimuli.

The molecular structure of the amphiphiles and their stereochemistry play a crucial role in the possibility of the formation of organized supramolecular nanostructures. Two common (but not essential) features usually encountered in the amphiphiles that self-assemble in organized nanostructures with high aspect ratios (that is *long and narrow, fiber-like objects*) are the presence of one or more chiral carbons and of moieties suitable for intermolecular directional interactions, such as hydrogen bonds (often secondary amide groups). In contrast, the interactions between the head groups in spherical micelles or vesicles have to be ascribed mainly to hydration and steric forces which do not have a structure directing power, resulting in a high rotational freedom of the monomers and low aspect ratios.

Self-assembled nanofibers are generally obtained by dispersing the amphiphiles into aqueous media by heating and/or sonication. Then the solution is allowed to stand at the designated temperature for a few minutes or several days, yielding the formation of the supramolecular assemblies.[6] In some cases, it is accompanied by macroscopic transformations of the solution such as the change of its rheological properties including the appearance of viscoelastic behaviour or gelation. In the latter case, the amphiphilic supramolecular assemblies become entangled, immobilizing the complete volume of water, which can thus support its own weight (often tested by turning the test tube upside down: if no flow is observed, the solution is said to have gelled).[7]

In the last three decades many researchers have designed and prepared a number of amphiphiles with very different molecular structures able to self-assemble into nanofibrillar structures in water. These promising supramolecular aggregates can be roughly classified into micellar-based aggregates and bilayer sheet-based aggregates.[6]

Micellar fibers. The simplest supramolecular aggregates with high aspect ratio formed by amphiphiles are micellar fibers that consist of material filled cylinders with the diameter of a molecular bilayer, when formed by single head/single tail amphiphiles, or of a single molecule length as in the case of bolaamphiphiles. A well-known example is the micellar fibrillar aggregates formed by some of the members of the *N*-alkylaldonamides deeply studied by Fuhrhop and coworkers. For example, the *N*-octylgluconamide **1** forms micellar rods that intertwine into quadruple helices as revealed by detailed image analysis of transmission electron (TEM) micrographs and solid state NMR spectroscopy (Fig. 3).[3] The growth and the stability along the long axis of the fibers have been explained by the cooperative formation of H-bond chains between the amide

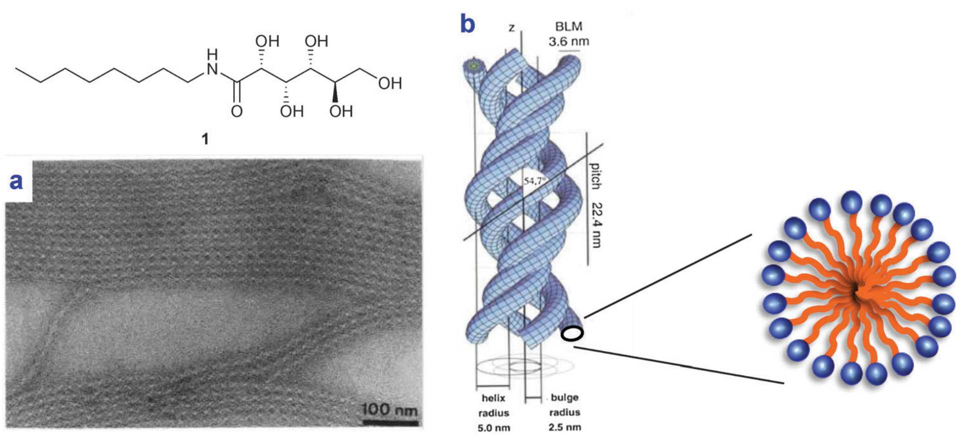


Fig. 3 (a) Electron micrograph of a two-dimensional array of *N*-octylgluconamide fibers, negatively stained with 2% phosphotungstate. (b) Computer graph of model structures of the quadruple helix. Adapted from ref. 3 with kind permission of The American Chemical Society.

groups, running parallel to the fiber surface. As a consequence, the micellar surface is not fluid but microcrystalline in nature (solid-like rods), differently from the fluid cylindrical micelles obtained by simply dissolving some amphiphiles in water above the Krafft temperature, in which non-directional repulsive forces between polar heads dominate. The chirality of the building blocks is expressed at the level of the supramolecular fibers. In fact, the D-enantiomer **1** forms left-handed helices while its L-enantiomer gives right-handed fibers. On the other hand, their racemic mixture does not form fibers, but precipitates as planar crystalline platelets upon cooling.[1,3,6]

Other kinds of amphiphiles have been found to self-assemble, under certain conditions, into micellar nanofibers, including *N*-acyl derivatives of natural amino acids and single tail peptide-amphiphiles. The latter contain a hydrophilic peptide sequence as a polar head connected *via* an amide bond to an aliphatic tail. In some rare cases, the cylindrical nanofibers intertwine to form multiple super helices analogously to **1**. An interesting example is the photoresponsive peptide amphiphile **2** studied by Stupp and coworkers, which self-assembles into a nanosized quadruple helix as revealed by TEM.[8] Upon irradiation at 350 nm a light-triggered dissociation of the super helix into single fibers was observed as a consequence of the photochemical cleavage of the bulky 2-nitrobenzyl substituent, which changes the molecular packing and the opportunity for interactions among molecules in neighbouring nanofibers (Fig. 4).

Another example is the double-stranded helix formed by the rationally designed sugar amphiphile **3**, containing an azobenzene moiety. The robust helical nanostructures result from cooperative multiple non-covalent interactions including: hydrophobic effect, π–π stacking and multiple H-bonds between the sugar heads.[9] In addition, these supramolecular nanohelices were found to be ''smart assemblies'' which can be shaped into spherical micelles, vesicles or unfolded micellar rods by external stimuli such as pH variation or light irradiation (Fig. 5).

As mentioned above, bolaamphiphiles can form monolayered micellar rods. This is the case of the α-amino-ω-lysine unsymmetrical bolamphiphile **4**, which forms long micellar fibers in water, with a diameter of 2 nm, corresponding exactly to the length of one molecule (Fig. 6).

Interestingly, it was found that chiral *trans*-1,2-bisureido cyclohexane-based bolaamphiphile **5** aggregates into segregated enantiomeric micellar rods in water, thus giving an amazing example of self-sorting based on chiral recognition. This was nicely demonstrated by using suitable fluorescence probes through the formation of exciplex and fluorescence resonance energy transfer (FRET) (Fig. 7).[10]

Bilayer sheet-based aggregates: helices, tubes and twisted ribbons. More interesting amphiphile assemblies are those based on distorted sheet-like bilayer membranes (single or multiple), with large curvature and a high aspect ratio, such as ribbons and tubes. Usually, they are formed by amphiphiles

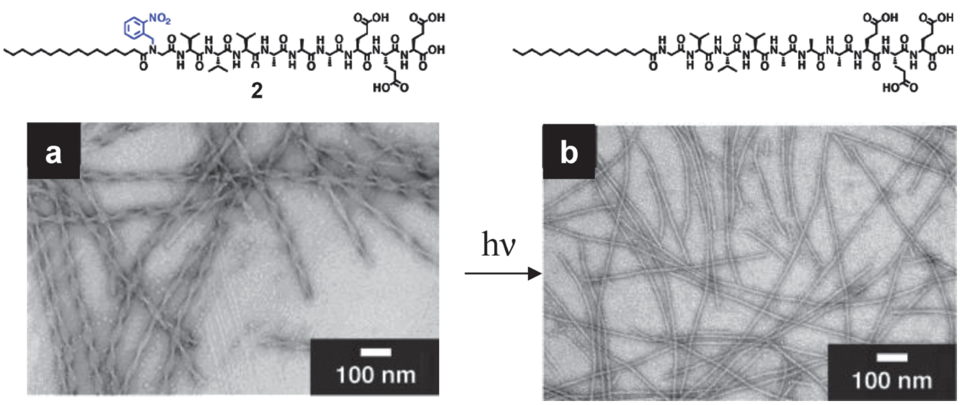


Fig. 4 TEM images of: (a) **2** annealed at 80 °C at a concentration of 7.4×10^{-4} M in pH 11 water containing NH_4OH for 30 min and slowly cooled to 25 °C over 90 min followed by air drying on the substrate. (b) Air-dried pH 11 water solution of photoirradiated (350 nm, 250 W, 5 min) **2** after annealing. Adapted from ref. 8 with kind permission of The American Chemical Society.

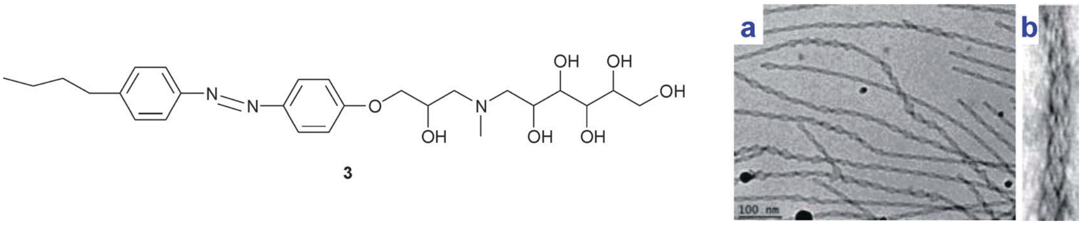


Fig. 5 (a) Cryo-TEM image of the double-stranded helices. (b) Enlarged picture showing the left-handed helicity of the double helix. Adapted from ref. 9 with kind permission of The Royal Society of Chemistry.

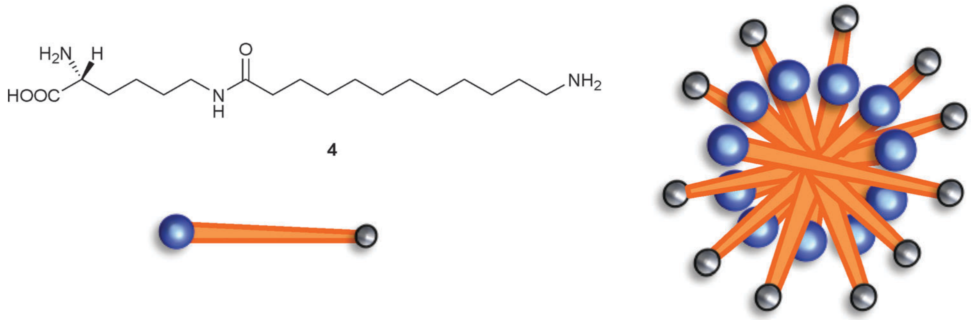


Fig. 6 Molecular structure of bolaamphiphile **4** forming one molecule thick micellar rods.

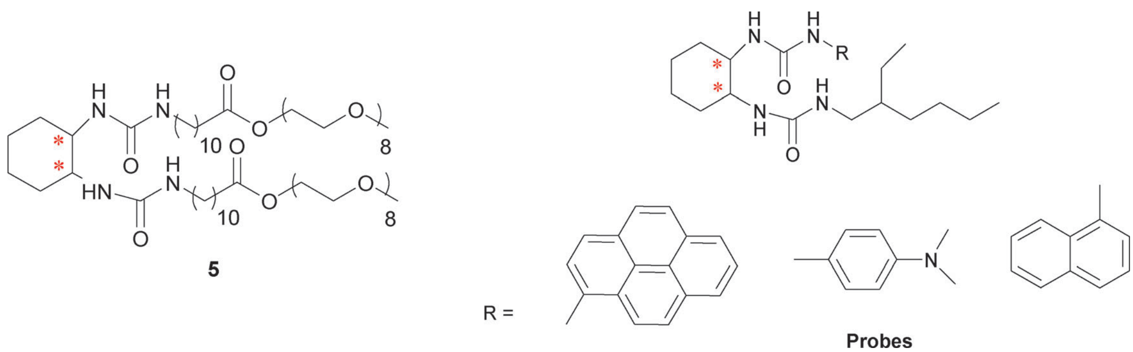


Fig. 7 Molecular structure of **5** and of the fluorescence probes used. Adapted from ref. 10 with kind permission of The American Chemical Society.

that are less soluble than those that form micellar rods. As stated above, natural phospholipids are known to form in water bilayer membrane structures, *i.e.* liposomes. In 1977, Kunitake *et al.* reported the first evidence of the spontaneous formation of bilayer vesicles from a totally synthetic dialkyl amphiphile.[6] Later, in 1984, the first evidence of the formation of high aspect ratio nanoassemblies in water from bilayer-forming amphiphiles came independently and simultaneously from three research groups in the United States and Japan.[11] As an example, Ihara *et al.* found that the double-chain lipid **6** with an oligo(L-glutamic acid) head group forms helical and tubular structures in water (Fig. 8). It is noteworthy that the thickness of the aggregates corresponds to that of single-walled bilayer structures.[12]

So far, a large number of amphiphiles forming fibrous aggregates based on distorted bilayers, also referred to as ribbons, have been reported. These include diacetylenic phospholipids, glycolipids with open or cyclic sugars and a different number and kind of unsaturation in the tail (double bonds, diacetylenic moieties), amino acid based amphiphiles as well as bolaamphiphiles and gemini surfactants. More detailed reviews can be found elsewhere.[3,6,7,12]

The ribbon-based aggregates can be morphologically classified into: (a) helical and tubular structures, featuring a cylindrical curvature, and (b) twisted ribbons with a Gaussian or saddle-like curvature[13] (Fig. 9a). In addition, experimental evidence on different systems has revealed that helical ribbons and tubules are intimately related. In fact, helices can be

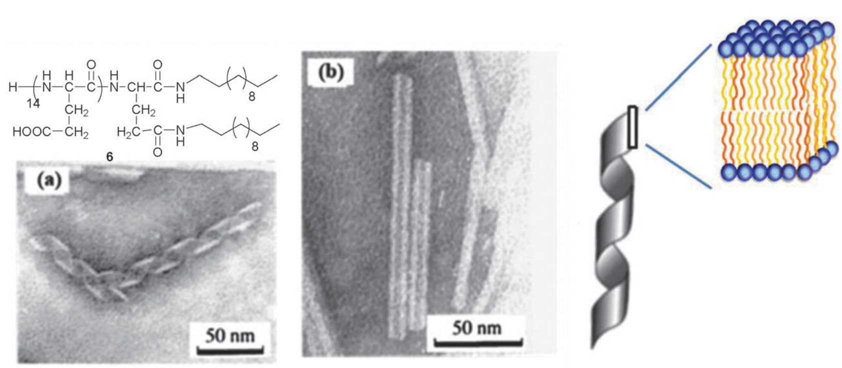


Fig. 8 TEM images of (a) helical and (b) tubular aggregates from lipid **6**. Adapted from ref. 12 with kind permission of Wiley VCH.

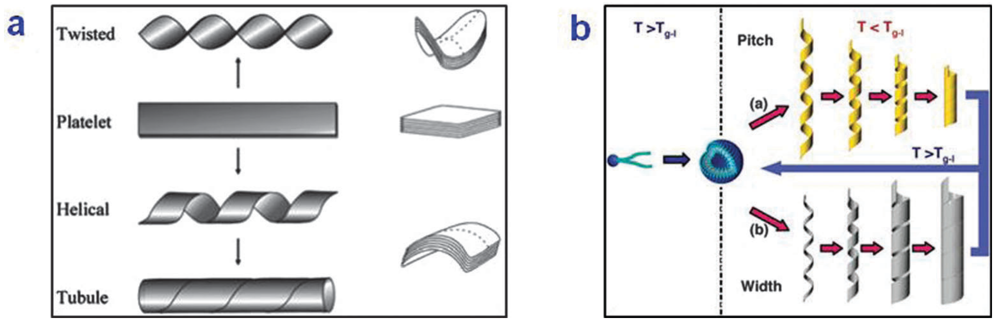


Fig. 9 (a) Schematic representation of twisted ribbons having saddle-like (Gaussian) curvature *versus* helical ribbons and tubules with cylindrical curvature. Adapted from ref. 13 with kind permission of The American Chemical Society. (b) Possible formation mechanism of amphiphile nanotubes from helical ribbons. Adapted from ref. 11 with kind permission of The American Chemical Society.

precursors of tubules under certain conditions and have to be envisaged rather as intermediates in tube formation. In particular, two mechanisms have been reported for the nanotube formation from helical ribbons (Fig. 9b): (i) shortening of the helical pitch of the ribbon at a constant tape width, and (ii) widening of the tape width at a constant helical pitch (more common). Both helices and tubes exist in solution only at temperatures below the gel-to-liquid crystalline phase transition temperature (T_{g-l}) of the amphiphile. Otherwise, at $T > T_{g-l}$ their morphological change to vesicles occurs.

To explain the formation of curved aggregates such as helices and twisted ribbons, different chirality-based theories have been used (chiral self-assembly model). In brief, chiral constraints force the molecules to pack at a nonzero angle with respect to their nearest neighbours (*i.e.* not parallel) generating a preferential orientation in the bilayer membrane. This induces twisting of the bilayer and yields the formation of hollow cylinders. The main drawback of such elastic theories is that they do not consider explicitly the presence of directional intermolecular interactions (hydrogen bonding, π–π stacking, *etc.*) that may be critical in the aggregate formation. Moreover, in the case of some bolaamphiphiles it was observed that the formation of nanotubes does not occur through the formation of chiral morphologies as twisted or coiled ribbons.

Control of nanofiber morphology. The ability to precisely control parameters such as the helical pitch or the inner and outer diameter of nanotubes and their multilamellarity is a key step towards their application in technology, materials and biological systems. As a result of a myriad of experimental observations, it is well known that both the molecular structure of the amphiphile and the experimental conditions in which the nanostructures are formed are crucial in determining the final outcome of the aggregation process.

In the series of the *N*-alkyl aldonamides investigated by Fuhrhop, the relative stereochemistry of the polyol head group (*i.e.* the nature of the head sugar) strongly affects the ability to form fibers as well as their morphology, ranging from micellar rods, to bilayer-based helical ribbons and rolled-up, cigar like, sheets. This is a consequence of the different molecular packing and hydrogen-bonding pattern induced by the different relative stereochemistry of the sugars.[1,3]

The number and the position of unsaturations in the hydrophobic portion was also found to strongly affect the self-assembly behaviour of synthetic amphiphiles into high-axial-ratio nanostructures. For example, Shimizu and coworkers reported striking different morphologies from the self-assembly of the series of phenyl glucosides **7–9**, which differ in the number of *cis* double bonds (1–3) in their hydrocarbon chain (Fig. 10).[11]

When amphiphiles contain ionizable groups, the aggregation process in water usually shows a strong dependence on the pH. Boettcher and coworkers reported an impressive example of pH modulation in the aggregation of the amino acid based amphiphile **10**.[14] Depending on the pH, multilamellar helical ribbons, multilamellar tubules or uniform tubes were observed as illustrated in Fig. 11.

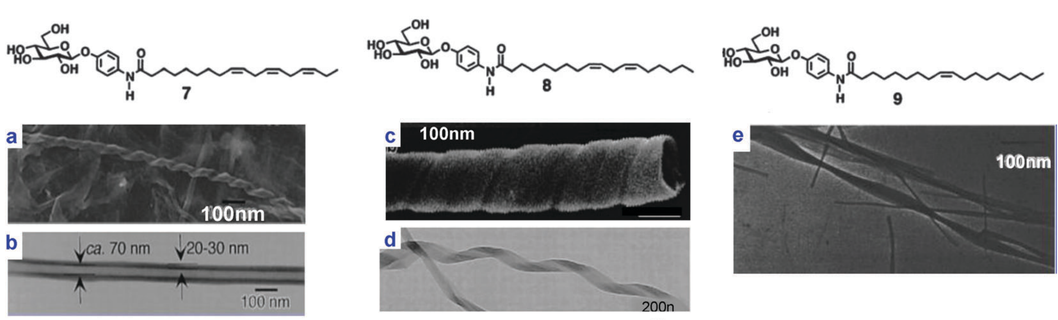


Fig. 10 EF-TEM and SEM images of the self-assembled morphologies of **7–9**. Adapted from ref. 11 with kind permission of The American Chemical Society.

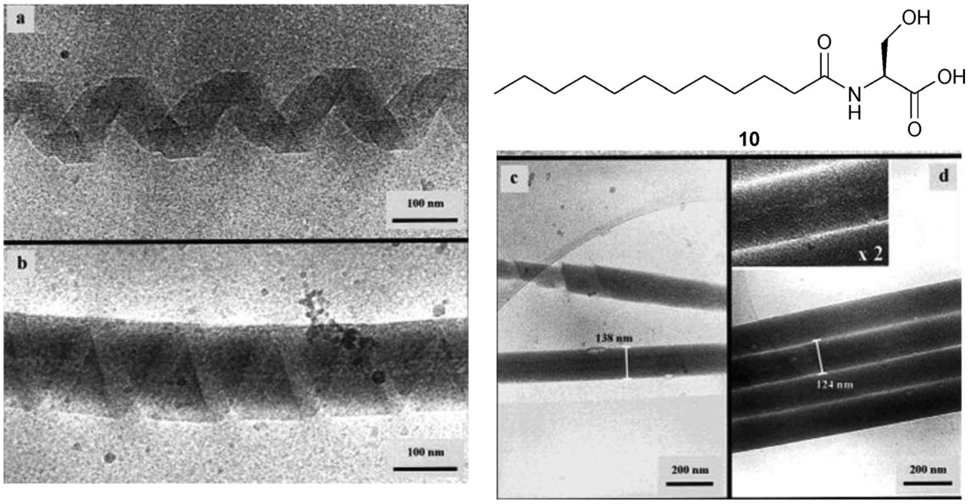


Fig. 11 Cryo-TEM of assemblies of amphiphile **10** in vitrified water (citrate buffer at pH 6.4): (a) multilamellar helical ribbons; (b) intermediate state of aggregation illustrating the growth process from ribbons to closed tubules; (c) multilamellar tubules in different states of aggregation; (d) smooth tubular rods of the same size at pH 4.9 (acetate buffer). Adapted from ref. 14 with kind permission of The American Chemical Society.

Beyond the critical role of molecular structure in determining the aggregate morphology, slight changes in the experimental conditions (concentration, temperature, cooling rate, ageing) may also have a dramatic effect on the self-assembly and thus on the nanostructure morphology. In fact, fiber formation is often a kinetically controlled (out of equilibrium) process and thus it depends on the experimental protocol used and generally speaking on the ''history'' of the sample. For example, the morphological transition of helical ribbons to tubules upon ageing is a well-known process.

Recently, Raghavan and co-workers were able to track in real-time, by light microscopy observation, the pathway of microtubules formation from a vesicle solution through helical ribbons intermediates, in aqueous mixtures of the single-tailed diacetylenic surfactant **11**, and geraniol cosurfactant (Fig. 12).[15] Upon diluting a concentrated gel of **11**/geraniol, 100 nm unilamellar vesicles are formed as detected by TEM and dynamic light scattering (they are too small for optical microscopy observations). After 1 hour, helical ribbons start to appear which evolve into smooth tubules *via* the ''growing width'' mechanism (Fig. 12).

Oda and coworkers have extensively investigated the self-assembly behaviour of the non-chiral bis-quaternary ammonium gemini surfactant **12** (named 16-2-16), bearing the chiral tartrate as the counter anion. They found that this system self-assembles in water to form twisted multilayered chiral ribbons and behaves as a hydrogelator. The handedness of the ribbons depends on the configuration of the tartrate anions, and remarkably their twist pitch could be modulated continuously from infinite (flat ribbons) to about 200 nm by varying the enantiomeric excess (ee) from 0 (racemate) to 1 (enantiopure tartrate).[7] Later they found that gemini **12** can also form helical

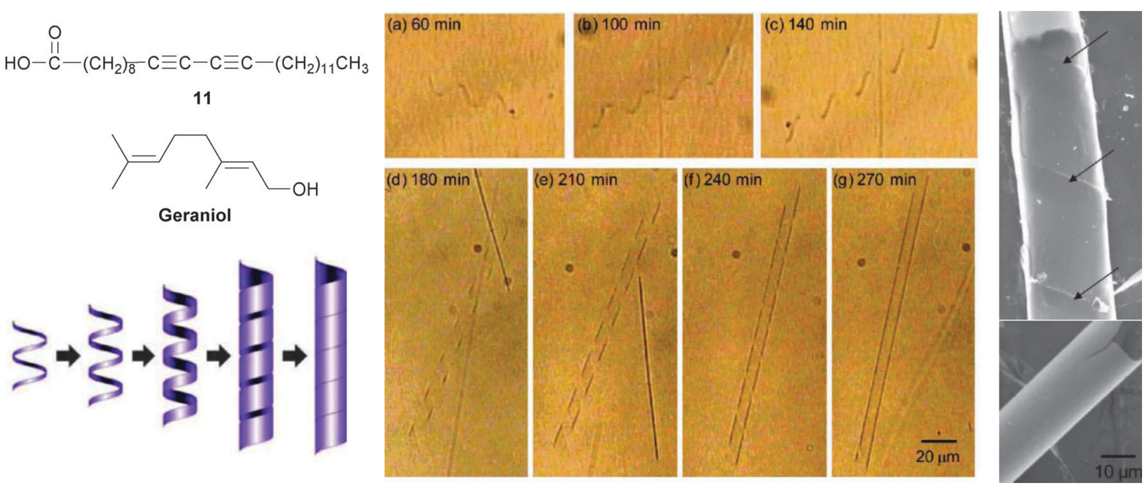
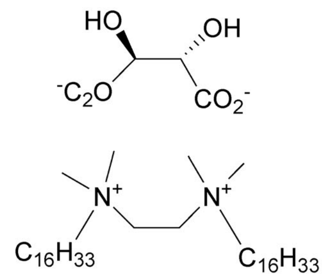


Fig. 12 Centre: optical micrographs in real time tracing the transformation of a helical ribbon to a closed hollow tubule at different times (through the ''growing width'' mechanism). Right: SEM images of the tubules with the overlap points indicated by arrows. Adapted from ref. 15 with kind permission of The American Chemical Society.



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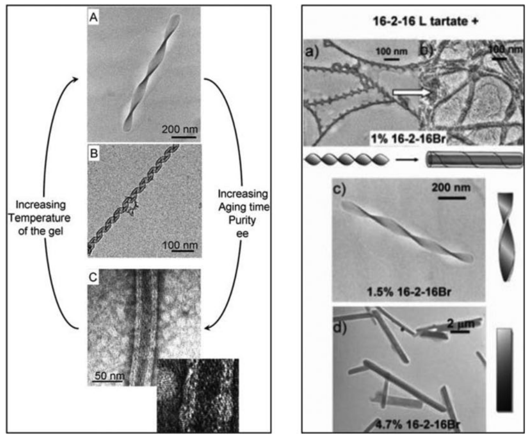
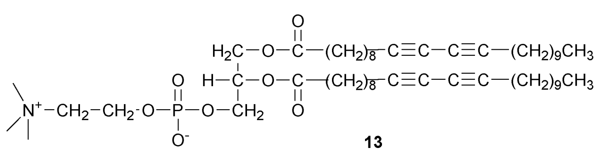


Fig. 13 Left: TEM images showing the morphological diversity of the self-assembly of gemini 16-2-16 tartrate surfactants: (A) twisted ribbon; (B) helical ribbons; (C) tubules. Upon variation of diverse independent parameters, the morphology can be fine-tuned. Adapted from ref. 16 with kind permission of The American Chemical Society. Right: TEM micrographs showing the effect of adding 16-2-16 Br to the morphologies of ribbons formed by 16-2-16 L-tartrate with ee = 1. (a) With 1% 16-2-16 Br, twisted ribbons transformed into helical ribbons and then into tubules after several days. (c) With 1.5% 16-2-16 Br, only twisted ribbons were observed. (d) Flat ribbons were observed with 4.7% added 16-2-16 Br. Adapted from ref. 13 with kind permission of The American Chemical Society.

ribbons and tubules and that the morphology and the dimensions of such assemblies can be finely tuned by varying external parameters such as temperature, ageing time, or slightly varying the amount of counter anion as schematized in Fig. 13.[16] For instance, the addition of a tiny amount of 16-2-16 Br to 16-2-16 tartrate **12** has a dramatic effect on the aggregate morphology and on its time evolution (Fig. 13).

Expression of supramolecular chirality. Molecular chirality seems to be intimately associated with the growth of self-assemblies with high aspect ratios and, consequently, with the formation of hydrogels. It is worth mentioning, for example, that almost all the amphiphilic hydrogelators are chiral. As mentioned above, when an amphiphile contains more than one stereogenic centre, their relative stereochemistry strongly affects their ability to form nanofibers, their morphology and, consequently, their gelation behaviour.[7] Racemates usually do not form fibers but rather the enantiomers tend to coaggregate yielding crystallization or precipitation in the form of flocs or platelets as in the case of amphiphile **1**.[3] However, some examples of self-sorting to form a racemic mixture of enantiomeric fibers (conglomerate) have been reported.[10]

An intriguing aspect is the connection between molecular chirality and fiber morphology, with regard to their possible application in functional nanodevices and as templates for inorganic nanostructures as described in Section 3. When chiral amphiphiles assemble into organized supramolecular structures, chirality is often, but not always, expressed in the morphology of these aggregates at a length scale of nanometers or micrometers.[7] In most of the cases, the handedness of the chiral self-assembly is determined by the absolute configuration of the monomer and the two enantiomers generate aggregates with the opposite handedness, as in the case of **1** (Fig. 3). Another clear example is the widely studied diacetylenic phospholipid **13**, *R*- and *S*-enantiomers of which form tubules possessing right- and left-handed helical traces, respectively (Fig. 14). In this case, the racemic mixture forms both right- and left-handed tubes, thus suggesting the separation of the two enantiomers in homochiral aggregates.[17] However, there are some exceptions to this simple picture. In fact, it has been reported that even a pure enantiomer or a non-chiral amphiphile, as in the case of lipids **14** and **15**, respectively, can give an almost equimolar mixture of right- and left-handed helices.[17]



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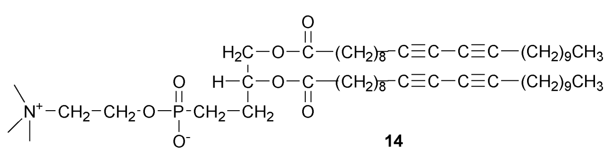

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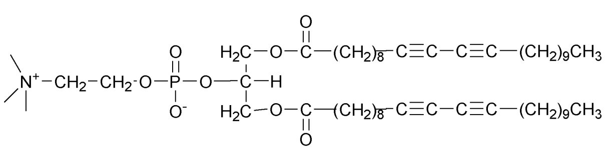

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Fig. 14 Molecular structure of diacetylenic lipids.

In these cases, it was suggested that the helix is formed by a statistical chiral symmetry-breaking mechanism rather than by a deterministic chiral-packing mechanism. Note that the aggregation of **15** in a racemic mixture of helices was the first example in which helical supramolecular structures were obtained by a non-chiral amphiphile, thus demonstrating that chirality, despite being a promoting factor, is not essential to helices and tubule formation.[17] It has to be stressed that fibers from **13–15** are formed in alcohol–water mixtures. Nevertheless, we have introduced these examples in the review because they are representative of more general concepts.

In our laboratory, we have recently developed new chiral amphiphiles based on polyfunctional cyclobutane platforms. The aggregation behaviour and the supramolecular chirality of their assemblies are currently under active investigation.[18]

To sum up, organized supramolecular nanostructures with fascinating shapes are formed from miscellaneous synthetic amphiphiles with a great structural variety. These supramolecular assemblies result from the cooperative effect of multiple weak interactions between a large number of components and are mainly kinetically controlled systems. For these reasons, the prediction and control over their morphology remain difficult tasks and clear design rules for the amphiphiles forming supramolecular nanostructures have not been defined. Nevertheless, experimental and theoretical approaches indicate important factors to fiber formation such as: (i) a moderate solubility into the aqueous media, (ii) intermolecular interaction moieties and their relative stereochemistry, (iii) molecular shapes suitable for an highly-ordered molecular packing and (iv) molecular chirality.[12]

2.2 Molecular and chiral recognition in amphiphile aggregates

Amphiphile aggregates such as micelles, liposomes and Langmuir monolayers are useful models to investigate molecular (and chiral) recognition in complex non-covalent polymolecular assemblies, which is another crucial aspect in supramolecular chemistry. In addition, they are simple models for biological membranes, so the investigations on the role of molecular structure, chirality and recognition events in the organization of such systems are often carried out with the additional aim of shedding light on the organization and functioning of biomembranes. The group of G. Mancini has widely investigated chiral recognition in biomembrane models, in particular micelles and liposomes, either from synthetic enantiopure amphiphiles as well as from natural lipids, exploiting suitable probes of chirality (Fig. 15). These investigations allowed them to highlight some interesting aspects.[19]

As an example, they found that chiral recognition in amphiphile aggregates can occur in the hydrophobic region, *i.e.* in sites far from the stereogenic centres of the chiral head groups, where enantioselectivity cannot be observed at the molecular level. This was the case of the recognition of the heterochiral pair of ditryptophan enantiomers (LD/DL) in micelles of surfactant **16** investigated by ^{1}H NMR spectroscopy (diastereomeric signals observed).[20] Other investigations made use of racemic (1 : 1) mixtures of rapidly interconverting enantiomers (*e.g.* biphenylic derivatives and bilirubin) as markers of chirality. In many cases it was found that the chiral recognition (deracemization) can be modulated by changing the experimental conditions. For example, in the interaction of the biphenylic derivative **19** with chiral micelles of surfactant **17**, it was found that the extent and the direction of the deracemization depend on the concentration ratio of two species and change reversibly upon pH variations.[21] On the other hand, the investigation of chiral recognition of bilirubin in micelles of surfactants **18** showed that the extent and the direction of the enantioselection depend on the length of the hydrophobic chain of the surfactant and on the concentration of the probe, the latter acting as a reversible switch for deracemization.[22] These examples clearly demonstrate that the expression of chirality in amphiphile self-assemblies cannot be straightforwardly correlated with the chiral information embedded in the monomer but it depends on the organization of the whole aggregate.

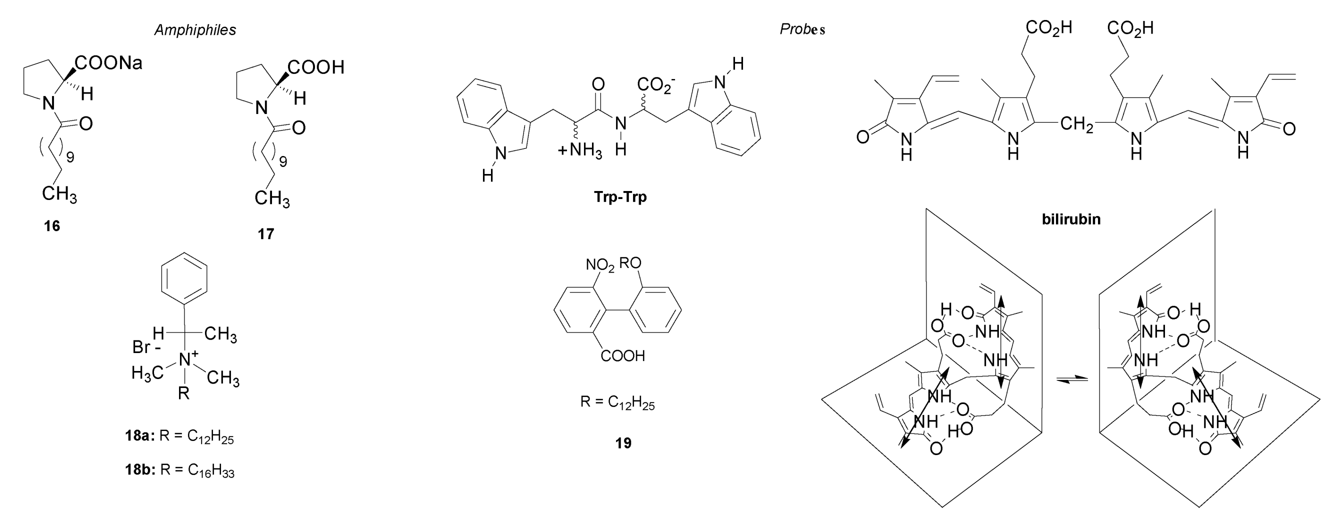


Fig. 15 Some amphiphiles and markers of chirality used to study chiral recognition in micellar aggregates.

Among the various amphiphile assemblies, Langmuir monolayers have revealed to be excellent models to study molecular and chiral recognition at the interface and molecular organization in two-dimensions. The unsurpassed advantage of the Langmuir technique in studying intermolecular interactions resides in the possibility of controlling the approach of the molecules to each other, and thus the degree and type of their packing, by varying the surface area. Chirality has been largely investigated in monolayers of a number of compounds, for example ester derivatives of *N*-acyl amino acid amphiphiles, using surface pressure *vs.* area per molecule (π-A) isotherm measurements, generally by comparing racemic and enantiomeric films. In addition, chirality, by affecting the microscopic organization of monolayers, controls not only their physicochemical behaviour but also their morphology. In fact, it is well known that mesoscopic domains featuring chiral shapes can be formed in monolayers in the condensed phase, as observable by various optical techniques, such as Brewster Angle Microscopy (BAM) and fluorescence microscopy, directly at the interface. Moreover, chirality of the underlying lattice structure of the monolayer has been detected by grazing incidence X-ray diffraction studies.[23] Thus, chiral recognition can be concluded on the basis of differences in the characteristic features of the π-A isotherms, different shapes of the condensed-phase domains or different lattice structures of the enantiomeric *vs.* racemic monolayers. As an example, it was observed that the enantiomeric monolayers of surfactant **20** are in a more condensed state than the racemic monolayers at the same molecular area, suggesting homochiral preference. In addition, the spiral-shaped domains of the two enantiomeric forms of **20** feature an opposite handedness, while the 1 : 1 racemic mixture tends to form compact domains with hooks of opposite curvature that are the signature of symmetry breaking. An even more spectacular example of chiral segregation has been observed comparing the condensed phase domains formed by the enantiopure and the racemic amphiphile **21** (Fig. 16).[23]

Chiral recognition of enantiomeric species such as odorants, cholesterol and surfactants has been investigated in monolayers formed by natural lipids. Interestingly, 1,2-dipalmitoyl-*sn-glycero*-3-phosphocoline L-DPPC monolayers are not able to discriminate the enantiomers of cholesterol, but are able to discriminate *R*-(−) and *S*-(+)-carvone as evidenced by *π*–*A* isotherms and theoretical studies. It is worth noting that the molecular recognition of the carvone enantiomers by the membrane lipids of the olfactory mucous is at the basis of their different smell, spearmint and caraway, respectively.

A number of investigations have focussed on the interfacial molecular recognition of non-surface-active *guest* substances dissolved in the aqueous subphase by *host*-monolayers of opportunely designed amphiphiles. Molecular recognition in these systems is based on complementary hydrogen-bonding, complexation and electrostatic interactions. Usually, the

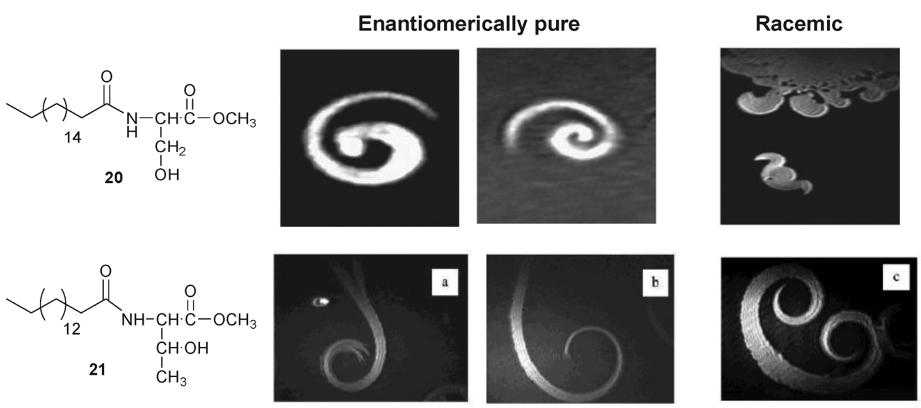


Fig. 16 BAM images showing chiral discrimination in condensed-phase domains in monolayers of amphiphiles **20** and **21**. Adapted from ref. 23 with kind permission of The American Chemical Society.

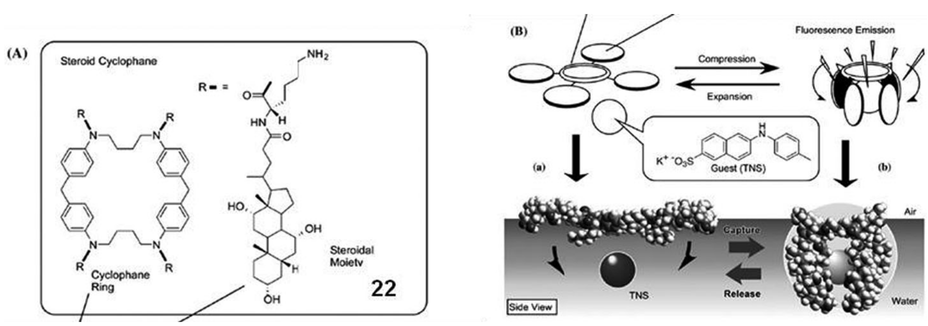


Fig. 17 (A) Molecular structure of the steroid cyclophane **22**. (B) Dynamic molecular recognition (capture) by the monolayer. Adapted from ref. 24 with kind permission of Elsevier.

presence of the non-surface-active guest changes drastically the shape of the π-A isotherm as well as the morphology and texture of the condensed phase domains as a consequence of the formation of a supramolecular complex between the two species at the interface.[24] Ariga *et al.* achieved dynamically controlled molecular recognition using the chiral steroid cyclophane **22** (Fig. 17). This amphiphile adopts an open conformation at low surface pressures, but upon compression of the monolayer a transition to a cavity conformation occurs. As a consequence of this conformational change, the reversible binding (capture and release) of an aqueous fluorescent guest was demonstrated.

3. Amphiphiles in nanochemistry and materials

3.1 Conventional amphiphile aggregates (micelles, rods, liquid crystals) used as templates for mesoporous silica and zeolite formation

The ability of amphiphiles to form self-organized aggregate structures and lyotropic liquid crystalline phases has been used to template the synthesis of nano- and mesostructured materials. Mesoporous materials have very large specific surface areas and well-defined mesoporous morphology and size (2 nm to 20 nm).

One of the most important cases of templating to synthesize mesoporous materials is that of amphiphile aggregates and silica units interacting in a non-covalent way, which makes the silica organize around the structure directing agent, in this case the supramolecular aggregate. The silica units then react with each other to generate mesostructured materials. Afterwards, the amphiphile is removed by calcination, which leaves a mesoporous material, usually harder and more stable because of this thermal treatment, with a templated pore structure. The removal of the template can also be done by solvent extraction. The whole procedure is also known as soft templating, where soft stands for the use of an organic material as a template (Fig. 18). In this section amphiphiles will be considered, although some biomacromolecules have been used for the same purpose too. This term is used in contrast to hard templating, where a rigid preformed porous object acts as a template for a material to adhere to it surface or to fill the void spaces. Combinations of both hard and soft templating have also been described.[25]

One of the key points of the process is to select the right template, in this case an amphiphile, to ensure that the desired nanostructure is formed and that the template can be easily removed.

Since the synthesis of the first families of mesostructured silicates by the scientists of Mobil, the MCM-41 silica and the M41S family,[26] using alkyl trimethylammonium salts, major improvements have been made in order to tailor the morphology and the shape of their pores (Fig. 19). These materials have potential applications in heterogeneous catalysis, in adsorption and separation processes, in host–guest chemistry and in drug delivery, amongst others.

The formation mechanisms. There are two main mechanisms for mesoporous silica soft templating (Fig. 20).[27] The first one implies the use of a micellar solution of amphiphile and is referred to as "cooperative self-assembly". In this mechanism there is a simultaneous assembly of the amphiphile and the silica precursor. Various mechanistic explanations have been proposed depending on which combination of charge of the amphiphile and the silica precursor is used, respectively, and whether an anion or a metal cation is involved.[28] The cooperative self-assembly can be summarized as follows: to a solution of the amphiphile in water at a certain pH, the silicate precursor is added. The usual procedure is to employ a simple organosilicate compound, such as TMOS (tetramethyl orthosilicate) or TEOS (tetraethyl orthosilicate) as the silica source, although some inorganic silica sources have also been described.

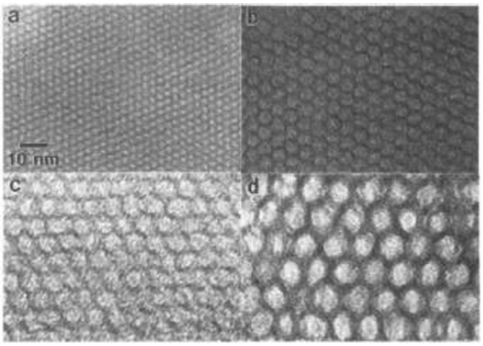


Fig. 19 Transmission electron micrographs of various MCM-41 silica materials of different pore sizes: (a) 20, (b) 40, (c) 65 and (d) 100 Å. Reproduced from ref. 26 with kind permission of The American Chemical Society.

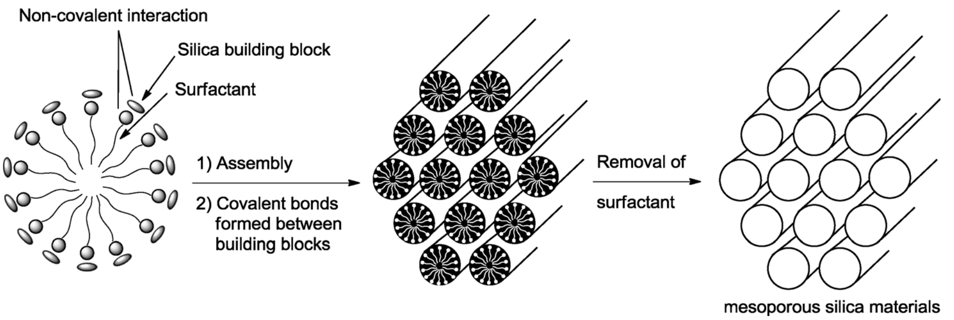


Fig. 18 Surfactant-directed formation of mesoporous materials.

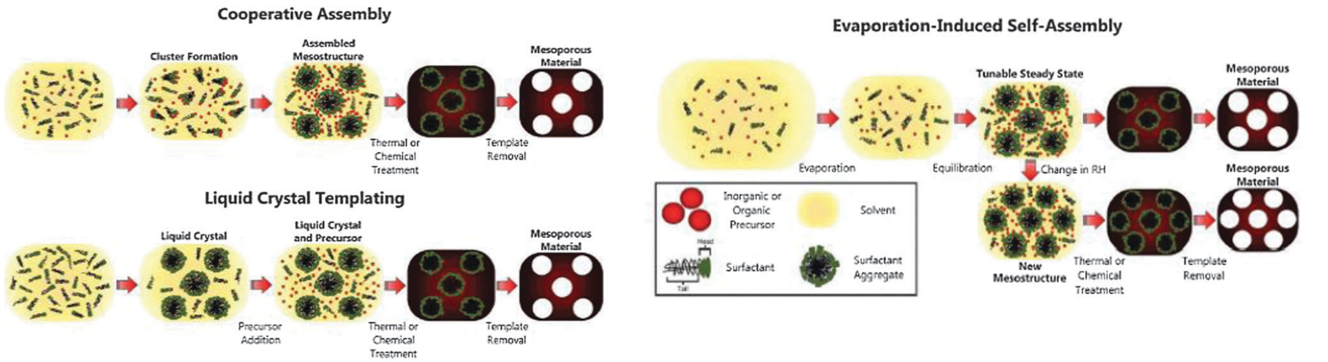


Fig. 20 Three main synthetic strategies for the synthesis of mesoporous materials: (a) cooperative assembly; (b) true liquid crystal templating (TLCT); (c) evaporation-induced self-assembly (EISA). Reproduced from ref. 25 with kind permission of The Royal Society of Chemistry.

The silicate precursors adsorb on the micelles. Then, the micelles associate into floes, where the condensation of the silicates takes place. This floes precipitate and, after elimination of the amphiphile, the mesoporous material is obtained. The symmetry, pore size and wall thickness of these materials are well controlled nowadays and are governed mainly by the amphiphile used. The second method implies the use of a liquid crystalline phase of the amphiphile, which means that a much more concentrated solution is used. This is also known as ''direct templating method'' or ''true liquid crystal templating (TLCT)''. In this method, a liquid crystalline phase is formed before condensation of the inorganic framework, the precursors of which are added later and not in the beginning. The silica precursors condensate in a very confined area around the amphiphiles, generally along with solvent evaporation. After removal of the amphiphile, the silica casts the mesostructure, pore size and symmetry of the liquid crystal scaffold.

A slight variation of this formation mechanism is the ''evaporation-induced self-assembly (EISA)''. This procedure is generally used to synthesize thin films of mesoporous materials. In this mechanism a volatile solvent is added along with the surfactant and the silica precursor. The mixture is spread over a large area and allowed to evaporate. A liquid crystalline phase is formed. Then, the silica precursor condensates around the liquid crystalline phase and the mesostructure is formed.

In all cases, removal of the surfactant can be achieved by chemical treatment or by thermal treatment. In many cases, thermal treatment implies shrinkage of the mesopores although some solutions have been found to avoid this problem.[25]

Most used amphiphiles[27]. Many examples of soft templating use cationic amphiphiles (**23–26** in Fig. 21), because most oxide surfaces at pH higher than their isoelectric points are negatively charged, and so they can interact with each other electrostatically. Most of them are of the type $C_nH_{2n+1}N^+(CH_3)_3X^-$ with n = 12–22 and X = Br or Cl. In general, the longer the alkyl chain, the bigger the mean pore diameter of the silica. The amphiphile counterion generally does not play an important role because similar structures have been observed for various counterions (*i.e.* Cl^- and Br^-). This is probably due to the fact that formation of the mesoporous material involves a cooperative interaction between the amphiphile and the growing silicate prepolymer, which displaces the counterion from the amphiphile aggregate. Cationic gemini surfactants (such as **27**, Fig. 21) have also been used. They are interesting because they self-assemble at a much lower concentration range and so less material is needed. Bolaform and multiheaded cationic amphiphiles (for example **28** and **29**, respectively, Fig. 21) have also been reported. Anionic amphiphiles have been less used to template mesostructured silica because generally there are no critical interactions between the surfactant head groups and the silica source. Nevertheless, there are some examples with polar heads that range from sulfates, sulfonates, carboxylates and phosphates (**30–32**, Fig. 21). In some cases, a co-structure directing agent (CSDA) is used to establish the appropriate interactions between the surfactants and the inorganic species, usually an amino-containing organoalcoxysilane (**33–34** in Fig. 21).[29]

Combinations of various amphiphiles have also been reported. Mixed amphiphilic systems such as cationic–cationic and cationic–non-ionic have been used to produce mesoporous materials. The best combination to control pore size is the cationic–anionic (catanionic) system.

Chiral amphiphiles for chiral mesoporous materials. Various chiral amphiphiles have been used in the soft templating of mesoporous silica and other related structures (Fig. 22). Some of them derive from natural amino acids and the chiral information is in the polar head (**35**). In other cases, the chiral information is far from the charged head of the amphiphile (**36–39**).

These chiral amphiphiles have been used to prepare mesoporous silicas that have chirality in their structure, in most occasions helical structures. For instance, the use of chiral sodium *N*-acyl-L-alanates together with a bit of the neutral amphiphile *N*-acyl-L-alanine and a CSDA such as **33** or **34** led to the achievement of one of the first mesoporous silicas with inherent chirality.[30,31] Specifically, twisted hexagonal rod-like morphologies were observed. The three components self-assemble to form a micelle that depending on the 3-component-mixture composition distorts leading to chiral structures. Co-condensation of the aminosilanes present (TMAPS or APS) with TEOS generated the chiral silica framework, the morphology of which was

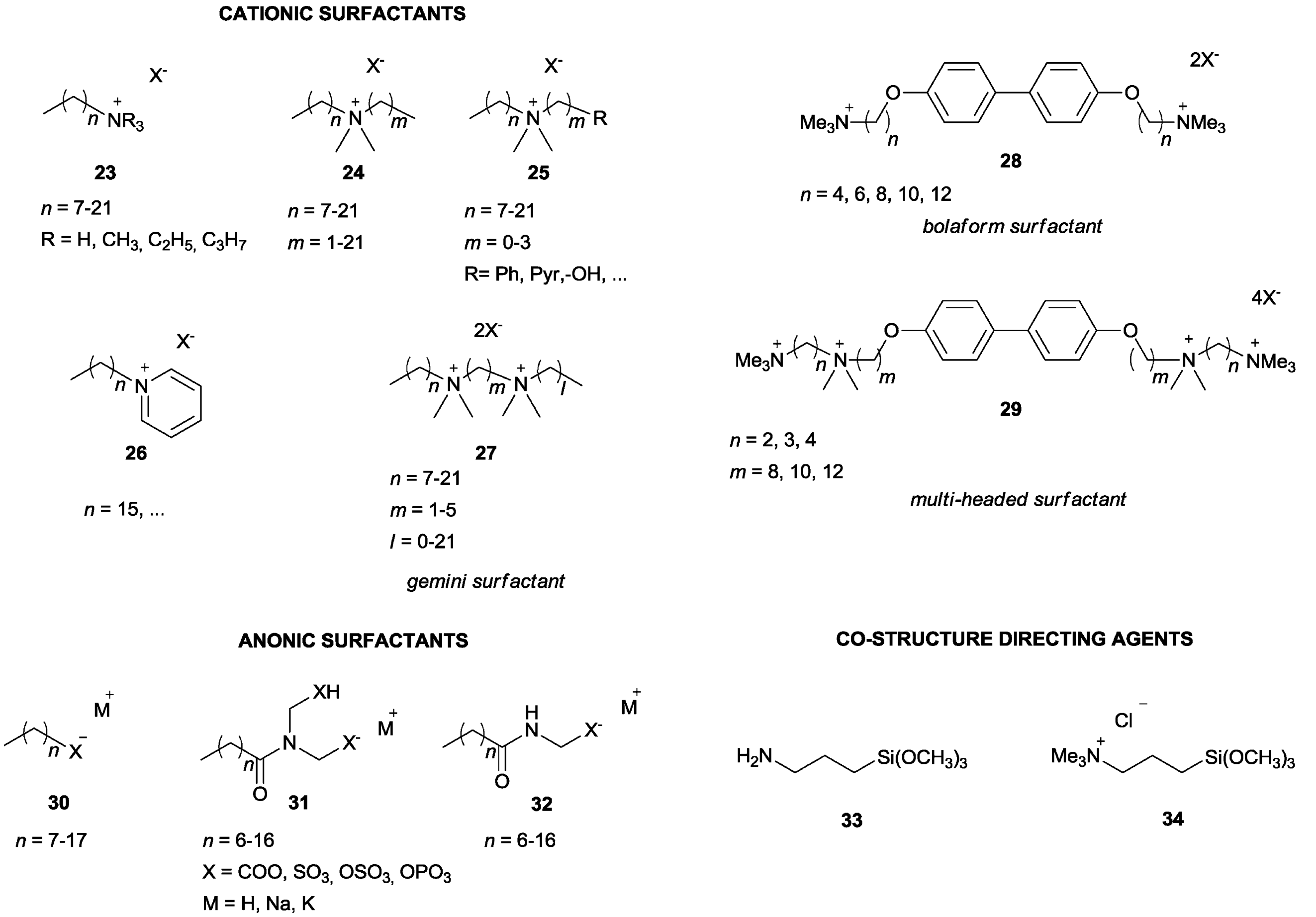


Fig. 21 Some of the most used cationic and anionic amphiphiles and CSDA in soft templating.

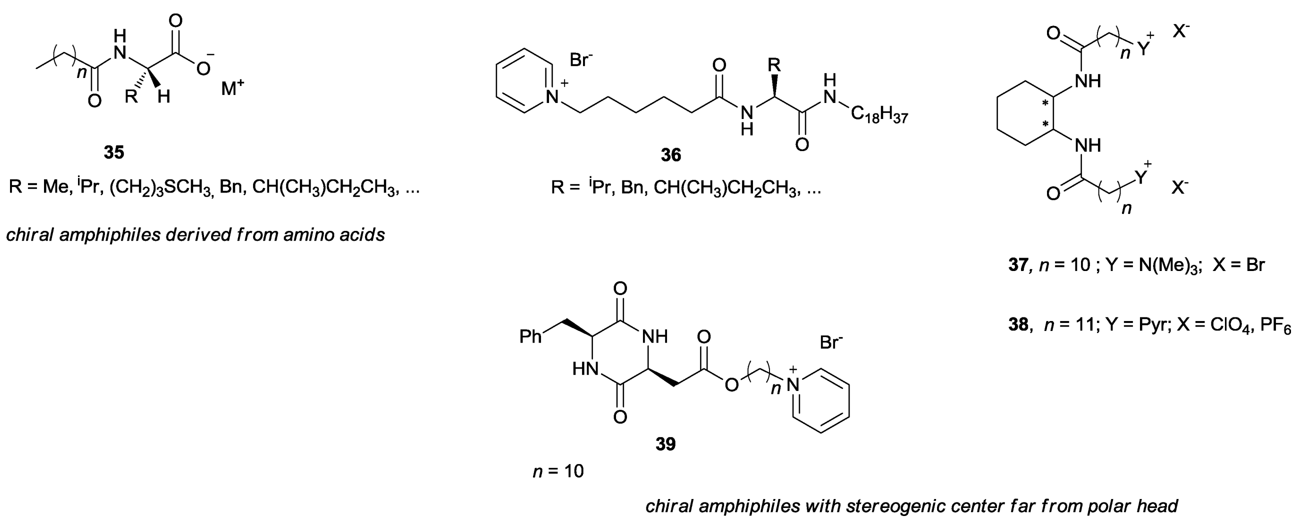


Fig. 22 Some of the most used chiral amphiphiles in soft templating.

confirmed by SEM and TEM (Fig. 23). These chiral rods contain chiral mesoporous channels inside, which were used in turn to synthesize Co and Pt nanowires. Since then, various other chiral amphiphiles have been used for the same purpose. The origin of the helical ordering of the mesostructure can be due to the helical packing of the chiral amphiphiles in the rod-like micelles, which in turn aggregate to give longer assemblies that adapt into helical forms because of the twisting power that originates from the helical packing (Fig. 23).[31] Other similar arrangements, mesoporous nanotubes with coiled pore

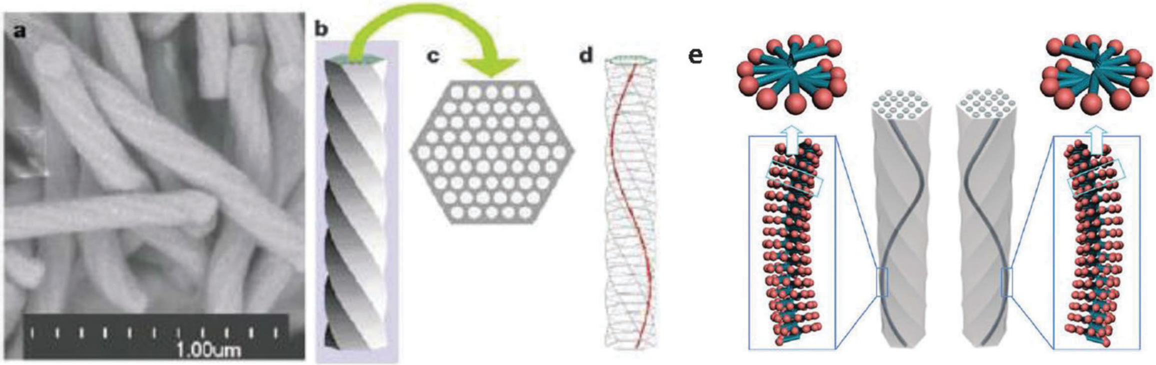


Fig. 23 (a) SEM images of the chiral mesoporous silica; (b) schematic drawing of structural model of the mesoporous material; (c) cross-section; (d) one of the chiral channels of the material. Reproduced from ref. 30 with kind permission of Nature Publishing Group. (e) Molecular origin of helical mesostructure derived from helical packing of chiral amphiphiles. Reproduced from ref. 31 with kind permission of The Royal Society of Chemistry.

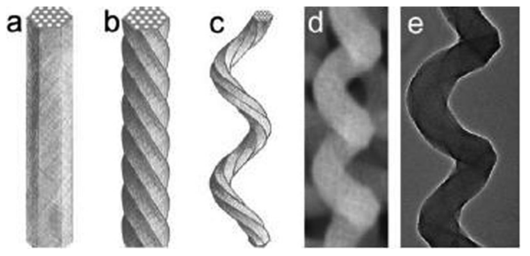


Fig. 24 Scheme of (a) a hexagonal mesoporous fiber; (b) a twisted hexagonal mesoporous fiber; (c) a spirally-curved twisted hexagonal mesoporous fiber (hierarchically helical structure); (d) SEM and (e) TEM images of hierarchically helical nanofiber. Reproduced from ref. 32 with kind permission of The Royal Society of Chemistry.

channels in the walls, have also been obtained using chiral cationic organogelators.

It is relevant to note that chiral mesostructured silica has also been obtained using achiral cationic and anionic templates,[31] thus demonstrating that molecular chirality is not the only factor governing helical construction. In some cases not only twisted hexagonal mesoporous fibers have been formed but also a second level of helicity, which is the result of a twist of the original already twisted mesoporous fiber (Fig. 24).[32]

It is noteworthy to mention that these materials are not only interesting because of their pores, but also because they show interesting microscopic morphologies. This is the case of controlled growth of CTACl (**23**, with n = 15, R = CH_3 and X = Cl) as template for TEOS, which gives mesoporous MCM-41 but with various different remarkable shapes as shown in Fig. 25.[33]

Mesostructured materials with zeolite type walls. Mesoporous zeolites can also be synthesized using the same strategies. In some cases, the temperature required for the synthesis has been a drawback. Moreover, it has not always been clear whether the structurally-directing amphiphile molecules are single molecules or micelles.[34]

It is not straightforward to obtain crystalline zeolites containing both micro- and mesoporous structures in a single phase if the aluminosilicate gel is directly crystallized in the presence of both ordinary organic mesopore-directing surfactants and molecular templates for the zeolite. Nevertheless, the use of surfactant-directed assembly of nanosized aluminosilicate precursors, which act like seeds and are building blocks that grow around the surfactant assemblies (generally, cationic ammonium based amphiphiles **23–25**), has been a successful strategy to overcome these difficulties.[35]

3.2 Amphiphiles which form ribbons, tubes, helices, used as templates for silica and other metal oxide nanostructures

Apart from the mesoporous silica that is synthesized by micelles or liquid crystal templation, some discrete objects with less common shapes can be achieved by the correct use of other supramolecular aggregates.[36] Some of these more complex inorganic structures have interesting applications: for example fibers and tubes can be used as nanowires in

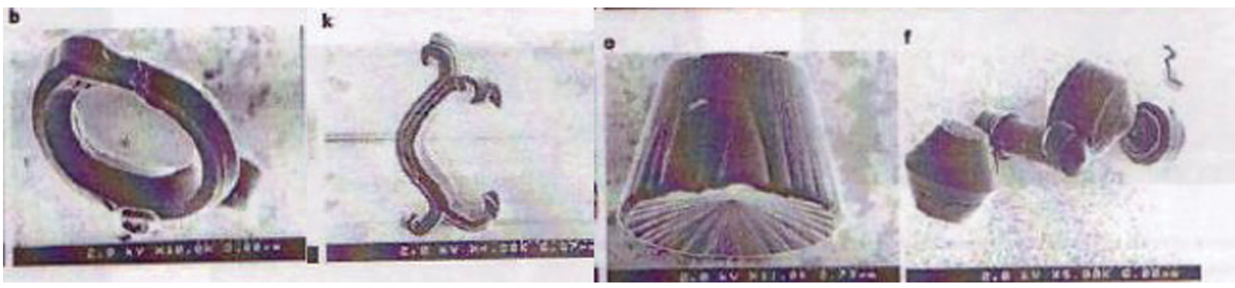


Fig. 25 Scanning electron microscopy images of representative mesoporous silica shapes. Adapted from ref. 33 with kind permission of Nature Publishing Group.

nanotechnology and optical devices, in drug delivery systems and in ion sensing; helices can be used as asymmetric reaction catalysts, as helical sensors and as optical devices; and hollow spheres in the controlled release of various molecules.[37]

Tubes. There are multiple possible mechanisms for their formation depending on which amphiphile and which material the nanotubes needed to be made of.[11,38]

For example, single silica nanotubes and bundles of them have been prepared starting with TEOS as the silica precursor and with amphiphiles such as laurylamine hydrochloride ($C_{12}H_{25}NH_2{\cdot}HCl$). Harada and Adachi revealed that the length of the nanotubes can be increased by introducing a quantity of trimethyl silane to the mixture during condensation of the gel and that the diameter of the nanotubes can be controlled by using a different length of the alkyl chain of the surfactant. The proposed mechanism is summarized as follows: the initial micelles are thought to become cylindrical along with the adsorption of TEOS. Then, these structures become short rods as the TEOS begins condensating. Afterwards, the short rods elongate to give nanotubes.[11]

Nanotubes have been prepared following similar procedures with other metal oxides, for example of vanadium oxide, in this case through a modified route where the surfactant head group is covalently linked to the vanadium oxide precursor;[39] or titanium oxide, taking advantage of the self-assembly of the amphiphile **38** ($X = PF_6$) and subsequent sol–gel polymerization of a titanium oxide precursor,[40] amongst others.

Helices. As shown in Section 2, some amphiphiles are good hydro- and/or organogelators. Some of these gels have been used as structure-directing agents to transcript the helical structures of the amphiphile aggregates to silica helices through the sol–gel process. Note that these are ''individual'' helices, and they are different from the helical mesoporous silica mentioned above.

A good example is the use of enantiopure *trans*-diaminocyclohexane-based cationic amphiphile **37**, mixed with some quantities of the directly related non-ionic organogelators. After the formation of the gel, polymerization of TEOS was carried out to obtain the formation of either right or left handed helices, depending on which enantiomer was used, which upon pyrolysis of the organic matter left hollow individual helices.[41] Note that in this case a mixture of organic solvent and water is required for the polymerization process. Intertwined chiral helical tubes of transition metal oxides (Ti, Ta and V) were synthesized following a very similar procedure to that mentioned above with amphiphile **38** with $X = ClO_4$ (Fig. 26).[42]

Another interesting example of the synthesis of silica single helices uses the aggregates formed by gemini amphiphile **12** shown in Fig. 13. Oda and co-workers studied the effect of various parameters, such as ageing time and temperature of the transcription process into inorganic materials observing interesting variations in morphology, including ribbons, helices and nanotubes (Fig. 27).[16] This example provides evidence that the ability to fine tune the morphology of the amphiphile aggregates allows the synthesis of inorganic materials with interesting nanostructures, thus revealing the potential of this approach.

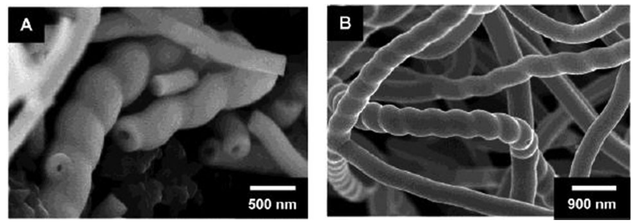


Fig. 26 SEM images of tantalum oxide fibers obtained from the (A) (*R*,*R*)-enantiomer and (B) the (*S*,*S*)-enantiomer of **38** with $X = ClO_4$. Reproduced from ref. 42 with kind permission of The American Chemical Society.

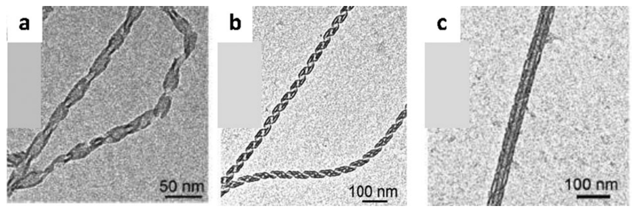


Fig. 27 TEM micrographs showing the morphology of silica (a) ribbons, (b) helices and (c) nanotubes obtained upon transcribing amphiphile **12** gels after various ageing times. Reproduced from ref. 16 with kind permission of The American Chemical Society.

Spheres and vesicles. Micelles and vesicles formed by some amphiphiles (such as octylamine and CTAB) have been used as templates to prepare hollow spheres and vesicles not only of silica but also of other metal oxides such as ZnO or Cu_2O.[37,43]

Apart from inorganic structures, silica and other metal oxides, ordered organosilicates, in which organic molecules are integrated into the walls of the material, can also be achieved by amphiphile templating methods such as the ones mentioned above. These periodic mesoporous organic–inorganic composites can be synthesized achieving well defined order and porosity too. Nevertheless, they have been left out of this tutorial review due to length constrictions.

4. Amphiphiles in biological and medicinal chemistry

4.1 Structure and function of bio-inspired amphiphiles

This section deals with self-assembled structures composed of amphiphiles where biological functionalities are embedded. Some of these assemblies mimic a variety of the functions of natural materials. Otherwise, conveniently functionalized amphiphiles produce assemblies that are good candidates for drug design.

The introduction of biological motifs based on peptides and carbohydrates into the structures of amphiphiles appears to be a useful avenue leading to a range of novel applications. In addition, because nucleic acids assume a range of tertiary structures that can have highly specific interactions with proteins, nucleic acid functionalized amphiphiles merit attention and will likely find potential applications in the areas of separation, drug delivery and biotechnology.

Papers dealing with peptide-based amphiphiles have reported progress in the rational design of molecules that self-assemble into predictable aggregate structures inspired by the structure of biological scaffolds. For instance, Tirrell *et al.* have demonstrated that amphiphile **40** tethered with collagen-like peptide sequences (Fig. 28) does spontaneously form a triple-helix in the peptide region thus mimicking the structural motif found in biological systems and further aggregates into spheroidal micelles.[44] This type of structure has already been mentioned in Section 2.

Another prominent field is the synthesis of bio-mimetic materials. In an effort to synthesize artificial materials that possess the complex structure and properties of bone, Stupp *et al.* synthesized a series of peptide-based amphiphiles that reversibly assemble into nanometer-sized fibers (Fig. 29). The process of assembly was triggered by addition of acid, divalent ions and oxidizing agents (leading to disulfide formation). The assembled bundles of amphiphiles successfully templated mineralization of hydroxyapatite ($Ca_{10}(PO)_6(OH)_2$) from $CaCl_2$ and Na_2HPO_4.[45]

Other examples can be found in studies of peptide amphiphiles as potential drugs. Chmielewshi *et al.* have reported advances in the use of peptide-based amphiphiles in drug design.[46] Fig. 30 provides an example of the type of structure under investigation where the amphiphile consists of a dimer of peptide that mimics the interface of HIV-1 protease. The two peptides are linked by an aliphatic chain. These peptide-based bola amphiphiles show high potency in inhibiting the protease.

Otherwise, the ability to combine bio-inspired strategies to build artificial molecular devices, with the possibility of imparting responsive and tuneable properties, typical of amphiphilic self-assemblies, has an impressive impact on nanoscience and soft matter science. Synthetic surfactants, where biological functional units (either as monomers or oligomers) are chemically inserted, are relevant. Among them, nucleosides, nucleotides, and peptidonucleic acids (PNA) derivatives can be found.

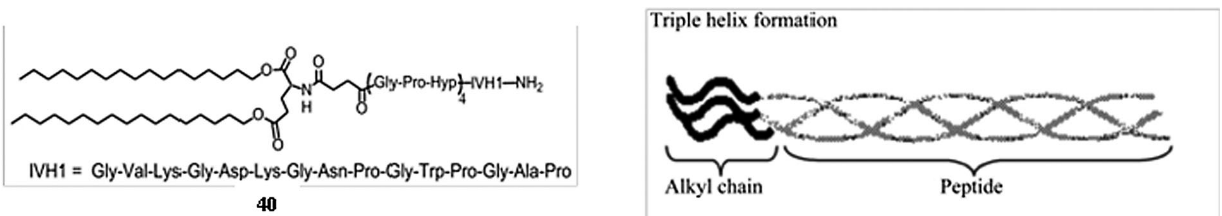


Fig. 28 Schematic representation of the triple-helix formed by double tail amphiphiles that contain the peptide sequence of type IV collagen. The structure and sequence of peptides in **40** is shown. Adapted from ref. 44 with kind permission of The American Chemical Society.

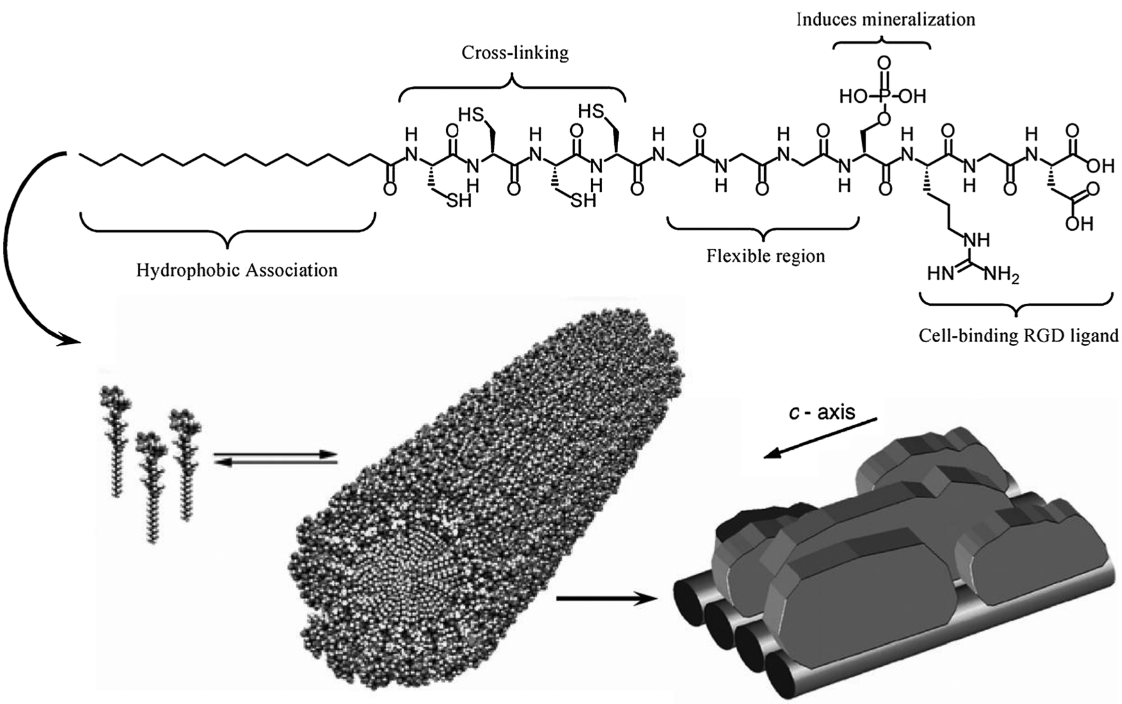


Fig. 29 Molecular structure of a peptide amphiphile that self-assembles into a cylindrical aggregate. The five key structural elements of the peptide are highlighted. The cylindrical aggregate assembles into fibres that template the mineralization of the hydroxapatite crystal. The *c* axis of the crystal is aligned with the long axis of the fiber. Reproduced from ref. 45 with kind permission of The American Association for the Advancement of Science.

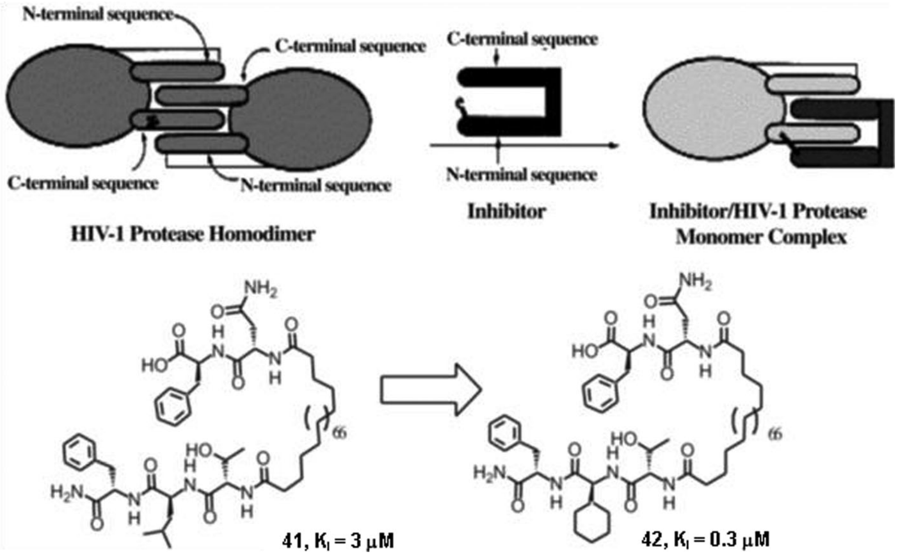


Fig. 30 Schematic illustration of the inhibition of HIV-1 protease homodimer by amphiphiles **41** and **42** that tether two peptides mimicking the interface of the protease. Optimization of the potency of the inhibitor is achieved by varying the residues on the amphiphiles, as shown below the illustration. Adapted from ref. 46 with kind permission of Wiley VCH.

While auto-organization of conventional surfactants represents the first generation of self-assembly, the second generation design of complex supramolecular structures is achieved by embedding chemical information units into amphiphilic polar heads, such as light or thermal responsive moieties, or, borrowing Nature's chemistry, nucleic acid bases. The next step, which is presently being explored, is the design of third generation self-assemblies, that is hybrids of such bio-inspired autoassociated systems with biopolymers (nucleic acids, proteins and so on).[47]

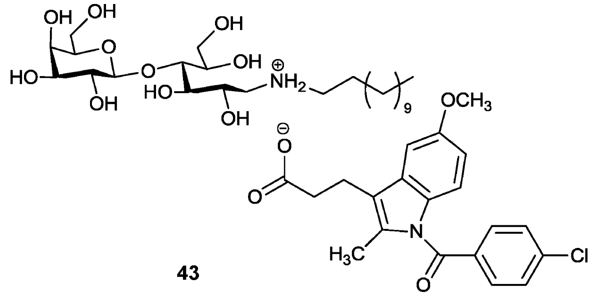
4.2 Drug delivery

Soft delivery systems have received increasing attention during the last few years. This is a result of many new candidate drugs displaying low aqueous solubility; hence a solubilising delivery system is often required for reaching sufficient drug bioavailability and/or facilitating clinical or even preclinical research and development work. Many of the soft delivery systems formed by surfactants may also be triggered to go from one structure to another by parameters such as temperature, ionic strength, calcium ions or other types of specific ions, pH, dilution with water, or the presence of specific metabolites. This facilitates storage and administration of the formulation in one form followed by transition to another after administration.

Aggregates of synthetic surfactants are used as soft drug delivery systems and nanostructured lipid carriers.[48] At the end of the 1960s, micelles attracted growing interest for application in drug vectorization. The anisotropic water molecule distribution in the structure of these objects allows the solubilisation of hydrophobic active principles in the micelles and enhances their bioavailability. In addition, active molecules are protected from enzymes that could degrade them and lead to their metabolism in biological media. The major drawback of micellar surfactant vectors is their tendency to break up upon dilution.[49]

Vesicles enable the encapsulation of larger amounts of drugs than matrix systems, which means that much smaller amounts of vectors can be administered. Most of these systems have versatile transport properties, and can vectorize hydrophilic and lipophilic substances. Among them, niosomes are bilayer vesicles made of nonionic surfactants. Niosomes are not sensitive to hydrolysis or oxidation and this is an advantage of their use in biological media. Moreover, surfactants are cheaper and easier to store than phospholipids. A further advantage of niosomes relative to liposomes lies in their formulation, as these vectors can be elaborated from a wide variety of surfactants, the hydrophilic heads of which can be chosen according to the application and the desired site of action.[49]

Another class of vesicles are formed by catanionic amphiphiles, which have not yet been widely used in vectorization. Nevertheless, these reservoir systems, which have versatile transport properties, can be formed from a great variety of surfactants and can be tuned according to the site of action where the drug has to be delivered by choosing an appropriate surfactant. For instance, Fig. 31 shows system **43**, which is involved in the cutaneous delivery of an anti-inflammatory drug



43

Fig. 31 Structure of a catanionic surfactant **43** from the association of an anti-inflammatory drug and a sugar-derived surfactant.

by direct association with a sugar-derived amphiphile forming a catanionic surfactant.

4.3 Biosurfactants

Biosurfactants, which are produced by microorganisms, are emerging as very promising tools due to their potential applications, among them, in molecular pharmaceutics and biomedicine. Biosurfactants can have a strong killing action on some types of cells, with lysis of red blood cells or fungal zoospores used as a bioassay. The interesting question, however, is whether more resistant cells, for example, bacteria with cell walls, may be killed by biosurfactants. At the present time, very few studies have been directed towards the possible wound healing properties of biosurfactants. This area of study certainly warrants further investigation and extension to other biosurfactant molecules because there would be a large market for a safe, cheap wound-healing additive for over-the-counter products.

Biosurfactants offer the possibility of replacing chemical surfactants, produced from nonrenewable resources, with alternatives produced from cheap renewable feed-stocks. They are also attractive because they are less damaging to the environment yet are robust enough for industrial use. The most promising biosurfactants at the present time are various kinds of glycolipids produced by *Candida yeasts*, *Pseudozyma yeasts*, or *Pseudomonas*. Despite the current enthusiasm for these compounds several residual problems remain and active efforts are being addressed to the imminent commercial exploitation of a new generation of microbial biosurfactants.

These compounds have now come to the top of the agenda of many companies as a result of the sustainability initiative and green agendas. Potential areas for use are expanding rapidly and useful outcomes will depend on whether biosurfactants can be tailored for specific applications, and whether they can be produced at a price that will make them attractive alternatives to chemical surfactants.[50]

5. Concluding remarks

In this review we have described the multimolecular assemblies formed in water by head(s)/tail(s) amphiphiles. These were reviewed from the point of view of supramolecular chemistry with a focus on the control of their morphology and chirality and on their recognition abilities. Their applications in nanochemistry, as templates for inorganic materials, as well as in biological and medicinal chemistry were also discussed.

Beyond the well known micelles and vesicles, amphiphiles can self-assemble in more organized high aspect ratio structures such as fibers, ribbons, helices and tubes, as driven by the hydrophobic effect and other structure directing interactions. The possibility to finely tune the aggregate morphology and chirality is crucial for their application in nanochemistry and materials chemistry. Both the molecular structure of the amphiphile (nature of the hydrophilic head and hydrophobic tail, stereochemistry, *etc.*) and the experimental conditions (pH, ageing time, *etc.*) have found to have a crucial role in determining the outcome of the aggregation. In some cases, the introduction of suitable groups in the amphiphile molecules allowed us to obtain responsive, ''smart'', aggregates in which the aggregation can be controlled by external stimuli. A particularly intriguing aspect is the relation between the molecular chirality and the supramolecular chirality and morphology of the amphiphile assemblies.

More common amphiphile aggregates such as micelles, liposomes and Langmuir monolayers revealed to be useful models for studying molecular and chiral recognition in complex polymolecular assemblies. In addition, they are useful biomembrane models. Intriguing phenomena such as the modulation of the recognition by changing the experimental conditions and chiral symmetry breaking have been reported.

Conventional amphiphile aggregates (micelles, rods and liquid crystals) have been used to template mesoporous silica materials and zeolites. In other cases, high aspect ratio amphiphile aggregates such as ribbons, tubes and helices have served as templates for discrete metal oxide and silica nanostructures.

Amphiphile assemblies in biological and medicinal chemistry are used as soft drug delivery systems. Moreover, bioinspired amphiphiles (for example, peptide-amphiphiles) and their aggregates have been investigated for the development of biomaterials and as potential drugs.

In summary, in this tutorial review we have offered an overview of the fascinating world, and potential applications, of amphiphiles in the 21st century with emphasis on supramolecular, materials and biological chemistry.

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